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12.	JTAG / PERIPHERALS
13.	ARDUINO HDRS
14.	SENSORS
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Revision History

Rev	Description	Date	Approved/Designer
X	Initial Schematic	JAN-2026	Efren D/ Antonio Q
A	Block diagram, add R806, update U11 MPN. Production Release	01/28/2026	Efren D/ Antonio Q

FRDM-A-S32K356

" QUETZAL "

Customer Evaluation Board

Notes:

- All components and board processes are to be ROHS compliant
- All connectors and headers are denoted Jx/JPx and are 2.54mm pitch unless otherwise stated
- All jumpers are denoted Jx. Jumpers are 2mm pitch
- Jumper default positions are shown in the schematics. For 3 way jumpers, default is always posn 1-2.
  2 Pin jumpers generally have the "source" on pin 1.
- All switches are denoted SWx
- All test points are denoted TPx

C A U T I O N :

This schematic is provided for reference purposes only. As such, NXP does not make any warranty, implied or otherwise, as to the suitability of circuit design or component selection (type or value) used in these schematics for hardware design using the NXP S32K family of Microprocessors. Customers using any part of these schematics as a basis for hardware design, do so at their own risk and Freescale does not assume any liability for such a hardware design.

Signals (ports) have not been routed via busses as this makes it harder to determine where each signal goes.

User notes are given throughtout the schematics.

Specific PCB LAYOUT notes are detailed in ITALICS

2 Different test points used in design:

TPx - Through Hole Pad small

TPHx - Through Hole Pad Large (for standard 0.1" header). Also used on IO Matrix (IOMx)



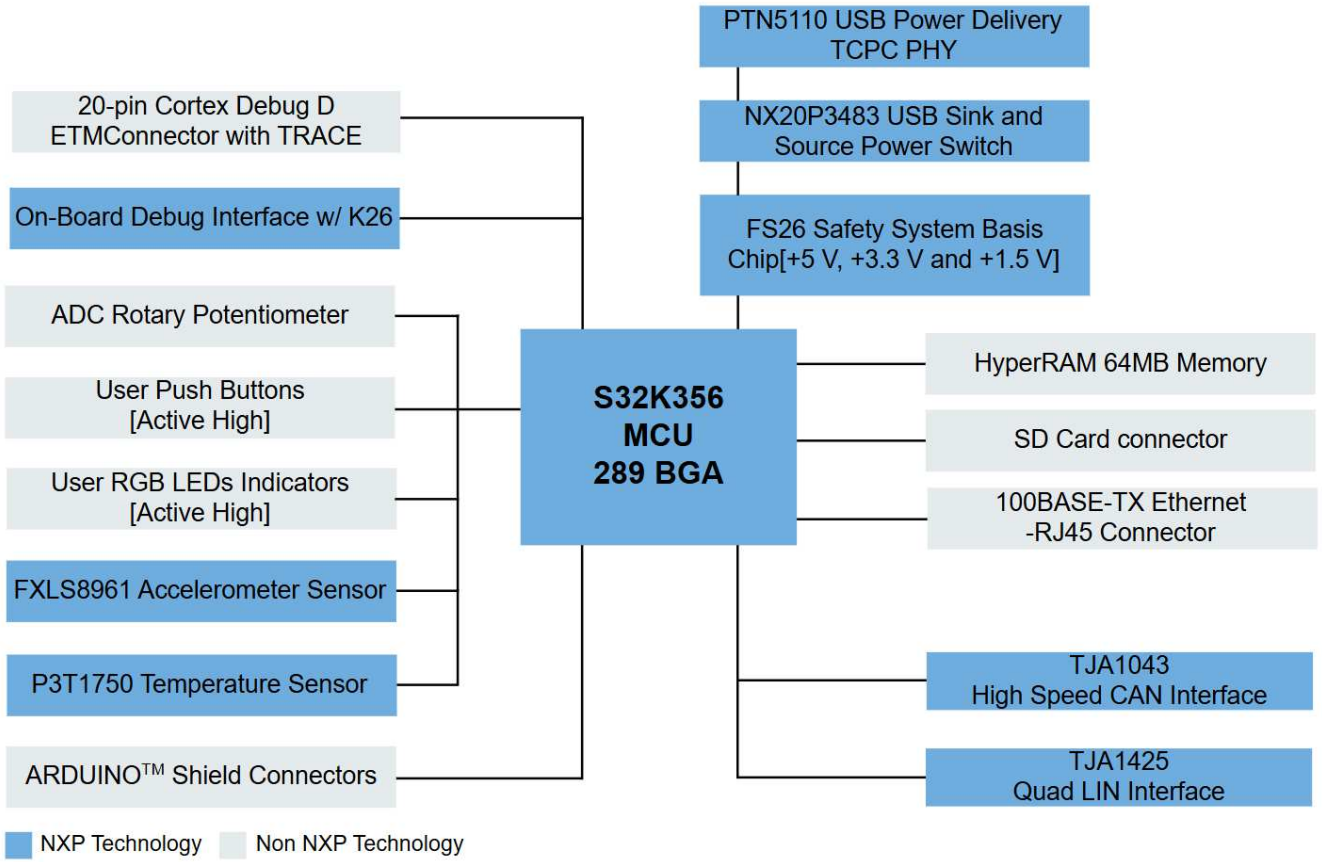
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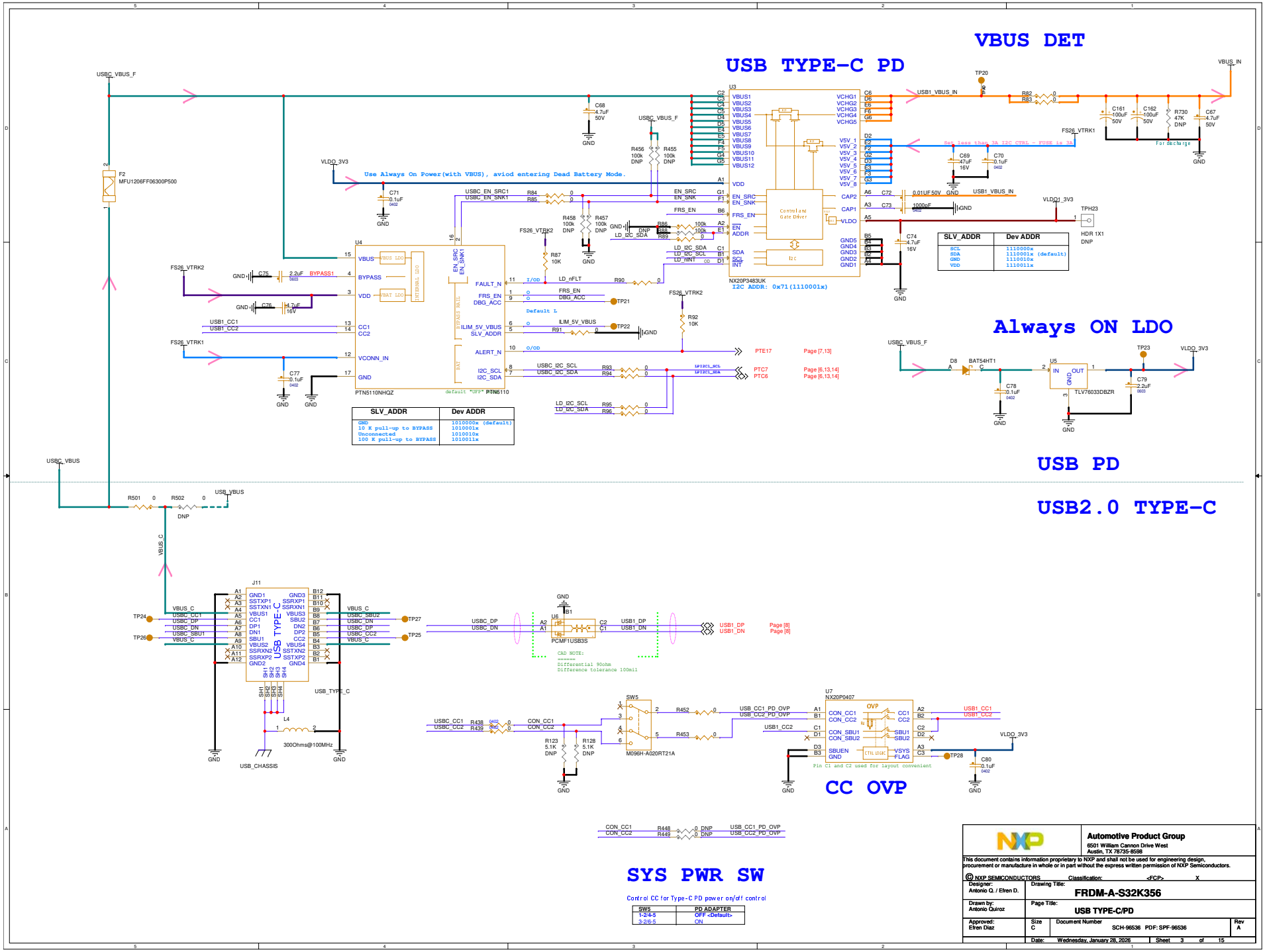
Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356	
Drawn by: Antonio Quiroz	Page Title: TITLE & NOTES	
Approved: Efren Diaz	Size C / Document Number SCH-96536 PDF: SPF-96536	Rev A
	Date: Wednesday, January 28, 2026	Sheet 1 of 15



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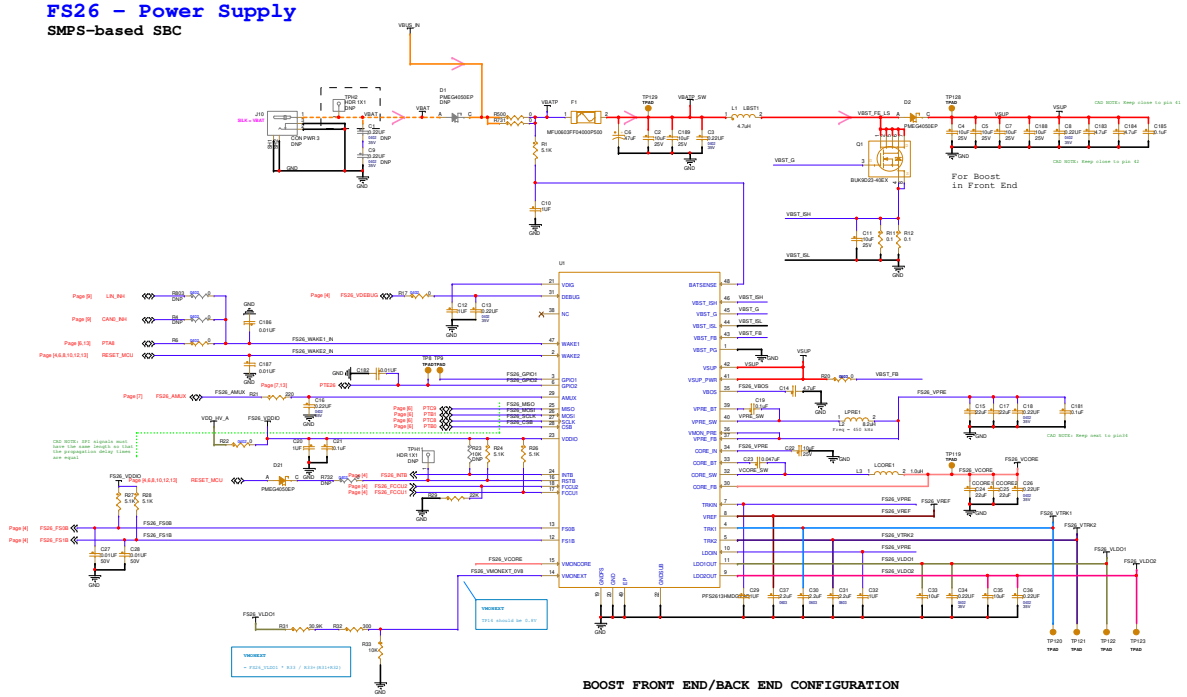
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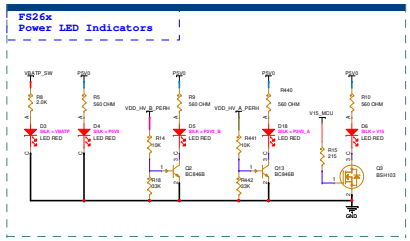
Automotive Product Group 6501 William Cannon Drive West Austin, TX 78735-8598	
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Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356
Drawn by: Antonio Quiroz	Page Title: BLOCK DIAGRAM
Approved: Efren Diaz	Size C; Document Number SCH-96536 PDF: SPF-96536; Rev A
	Date: Wednesday, January 28, 2026; Sheet 2 of 15

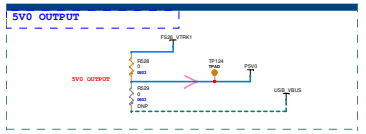

VBUS DET
USB TYPE-C PD
Always ON LDO
USB PD
USB2.0 TYPE-C
CC OVP
SYS PWR SW
USB TYPE-C PD
Use Always On Power(with VBUS), avoid entering Dead Battery Mode.
I2C ADDR: 0x71(1110001x)
PTN5110
NX20P3483UK
NX20P0407
TLV76033DBZR
PCMF1USB3S
M096H-A020RT21A
Control and Gate Driver
Pin C1 and C2 used for layout convenient
Control CC for Type-C PD power on/off control
SW5 | PD ADAPTER
1-2/4-5 | OFF <Default>
3-2/6-5 | ON
SLV_ADDR | Dev ADDR
1010000x (default)
1010001x
1010010x
1010011x
1110000x
1110001x (default)
1110010x
1110011x
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Designer: Antonio Q. / Efren D.
Drawing Title: FRDM-A-S32K356
Drawn by: Antonio Quiroz
Page Title: USB TYPE-C/PD
Approved: Efren Diaz
Size C
Document Number SCH-96536 PDF: SPF-96536
Rev A
Date: Wednesday, January 28, 2026
Sheet 3 of 15

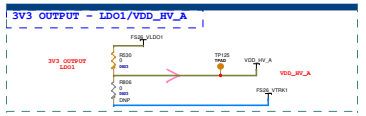
FS26 – Power Supply

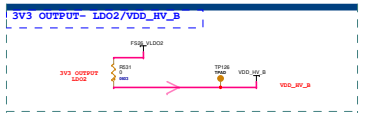
SMPS-based SBC

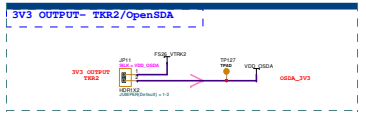


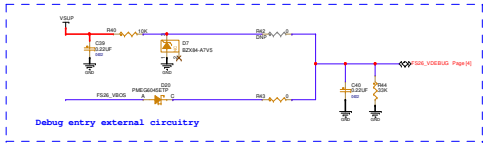












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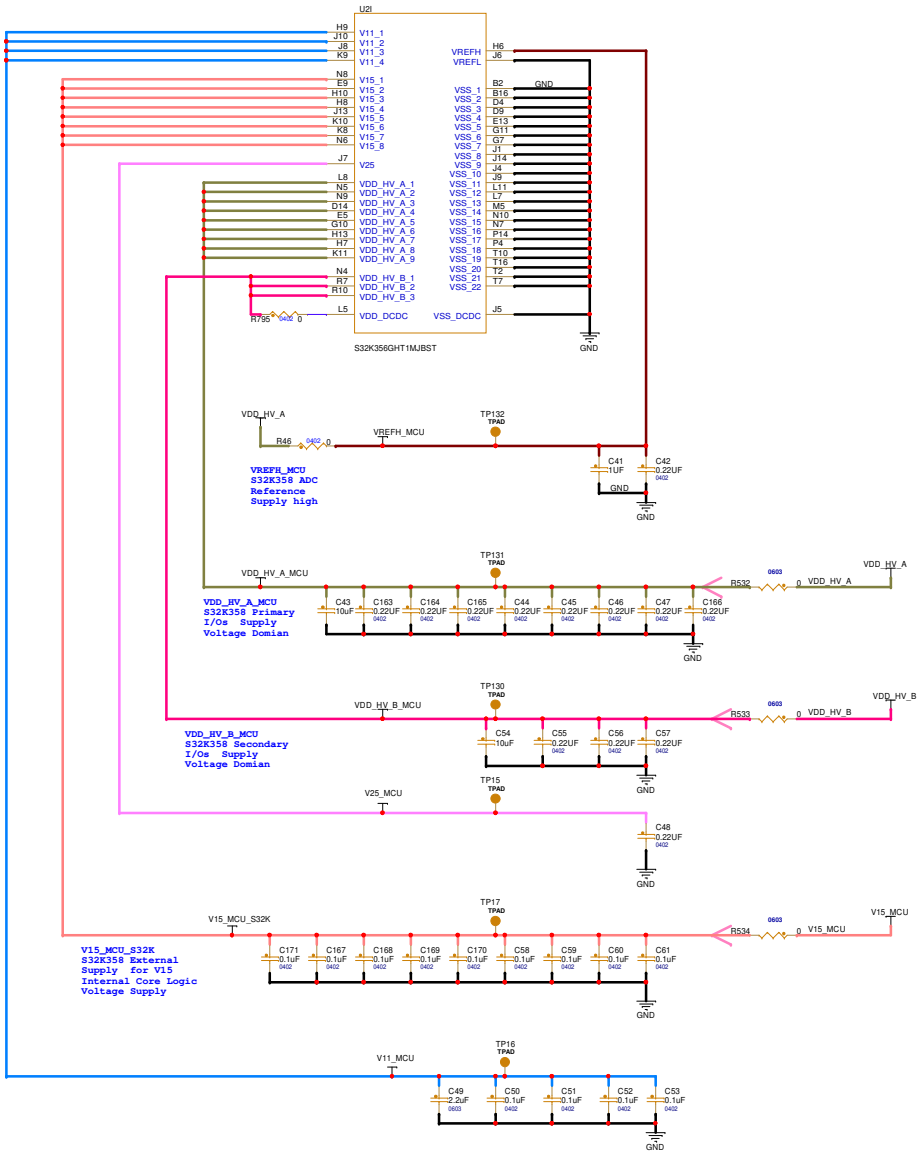
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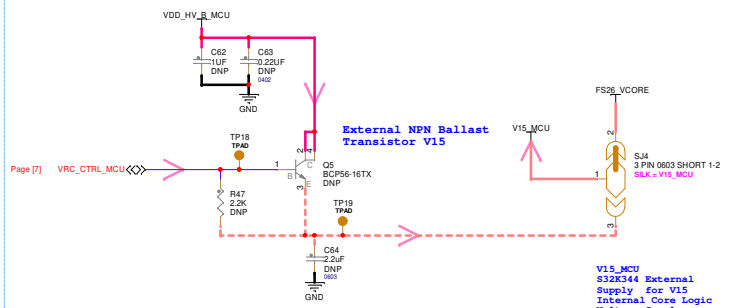
Designer: Antonio G. / Efren D.	Reviewing Title: FRDM-A-S32K356
Drawn by: Antonio Cuevo	Page Title: FS26 SBC Power Supply
Approved: Efren Diaz	Size: D; Document Number: SCH-96536 PDF:SPF-96536; Rev: A
	Date: Wednesday, January 28, 2026; Sheet 4 of 15

S32K35X Microcontroller – 289 BGA

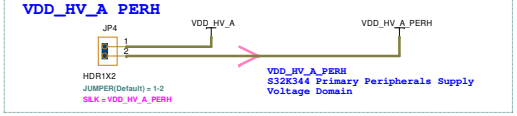
Power Supply Pins



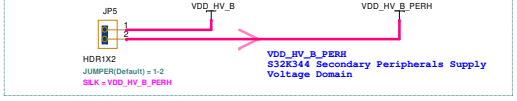
External NPN Ballast Transistor V15



VDD_HV_A PERH



VDD_HV_B PERH



TPH3
HDR 1X1
DNP
SILK = GND

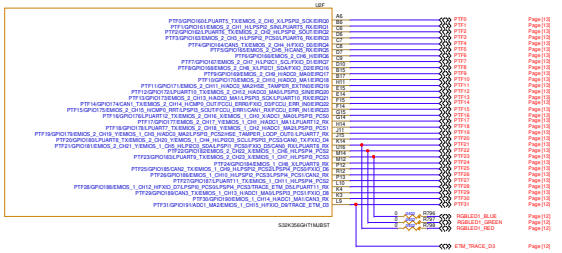
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Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356	
Drawn by: Antonio Quiroz	Page Title: S32K3XX - MCU Power Supply Pins	
Approved: Efren Diaz	Size C / Document Number SCH-96536 PDF: SPF-96536	Rev A
	Date: Wednesday, January 28, 2026	Sheet 5 of 15

S32K35X Microcontroller – 289 BGA

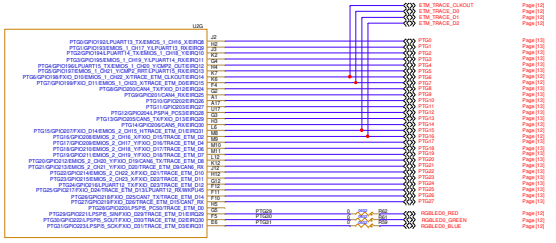
Ports A/B/C



Ports F/G





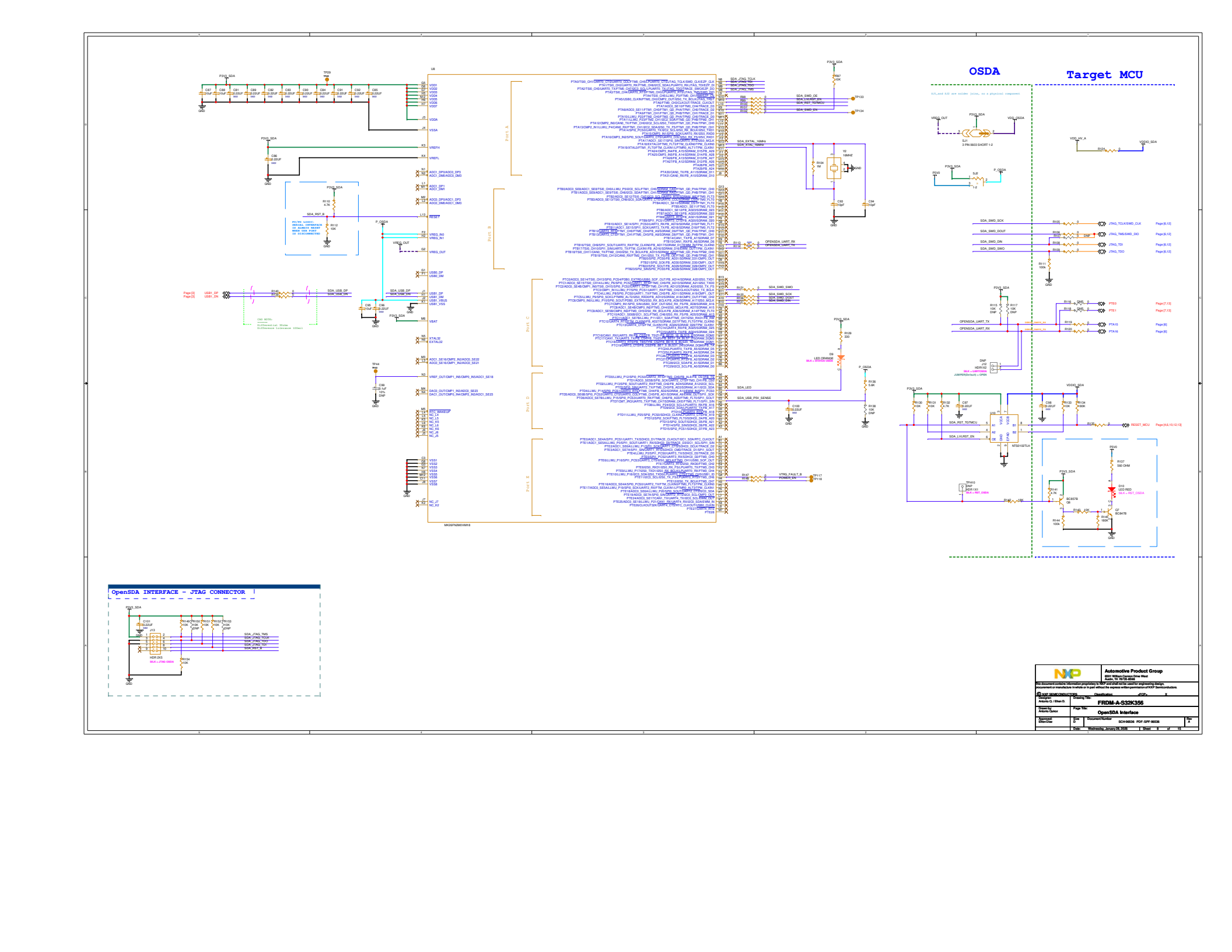


S32K35X Microcontroller – 289 BGA

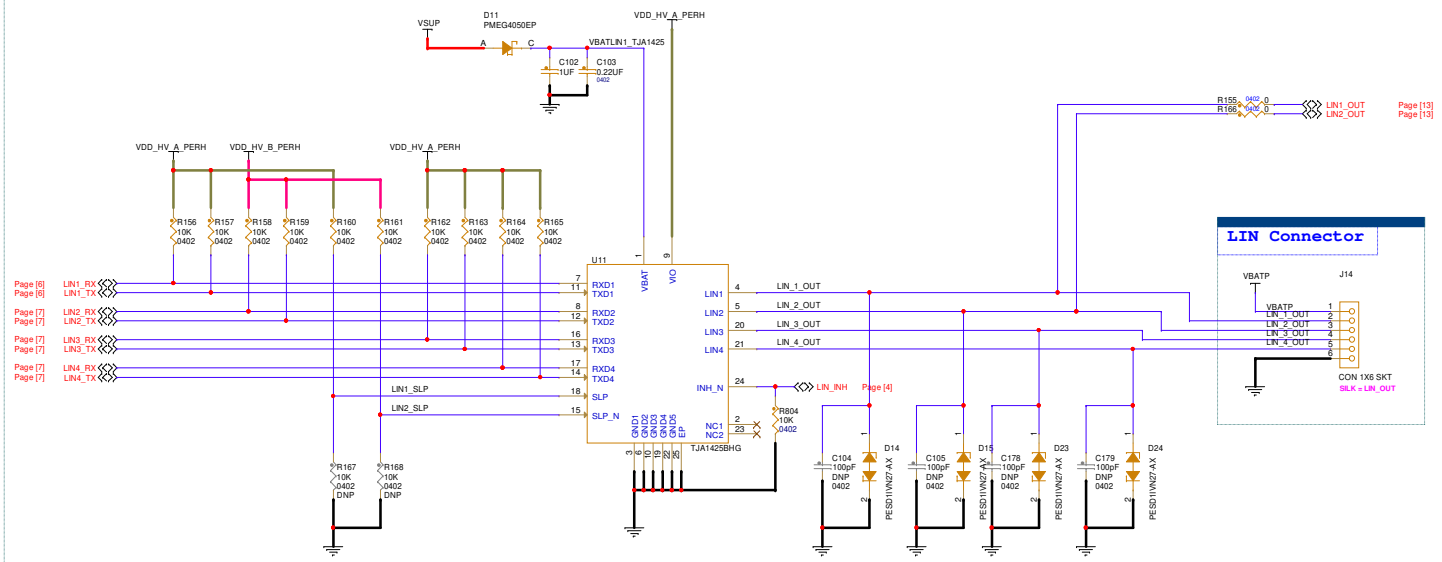
Ports D/E/H



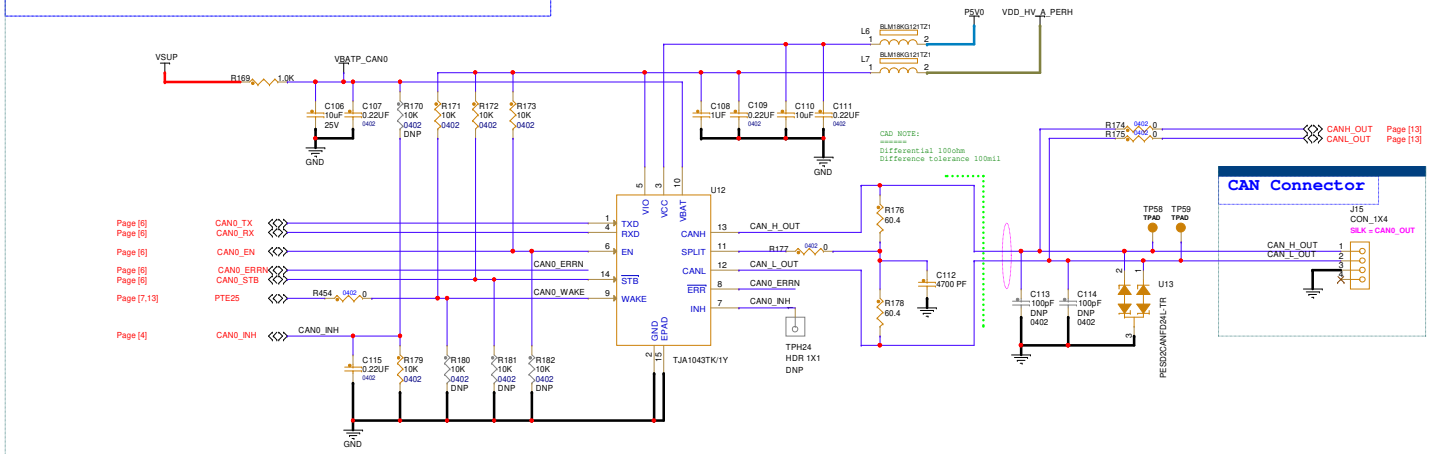
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Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356	
Drawn by: Antonio Quiroz	Page Title: S32K3XX - Ports D/E/H	
Approved: Efren Diaz	Size C / Document Number SCH-96536 PDF: SPF-96536	Rev A
	Date: Wednesday, January 28, 2026	Sheet 7 of 15


OSDA
Target MCU
OpenSDA INTERFACE - JTAG CONNECTOR
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FRDM-A-S32K356
OpenSDA Interface
Designer: Antonio G. / Efren D.
Drawn by: Antonio Cuevo
Approved: Efren Diaz
Page Title:
Size: D
Document Number: SCH-96536
PDF: SPF-96536
Rev: A
Date: Wednesday, January 28, 2026
Sheet 8 of 15

LIN Physical Layer
TJA1425BHG transceiver



TJA1043TK/1Y HS-CAN Transceiver



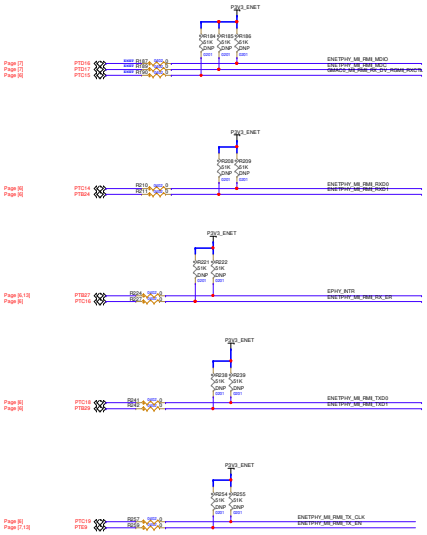
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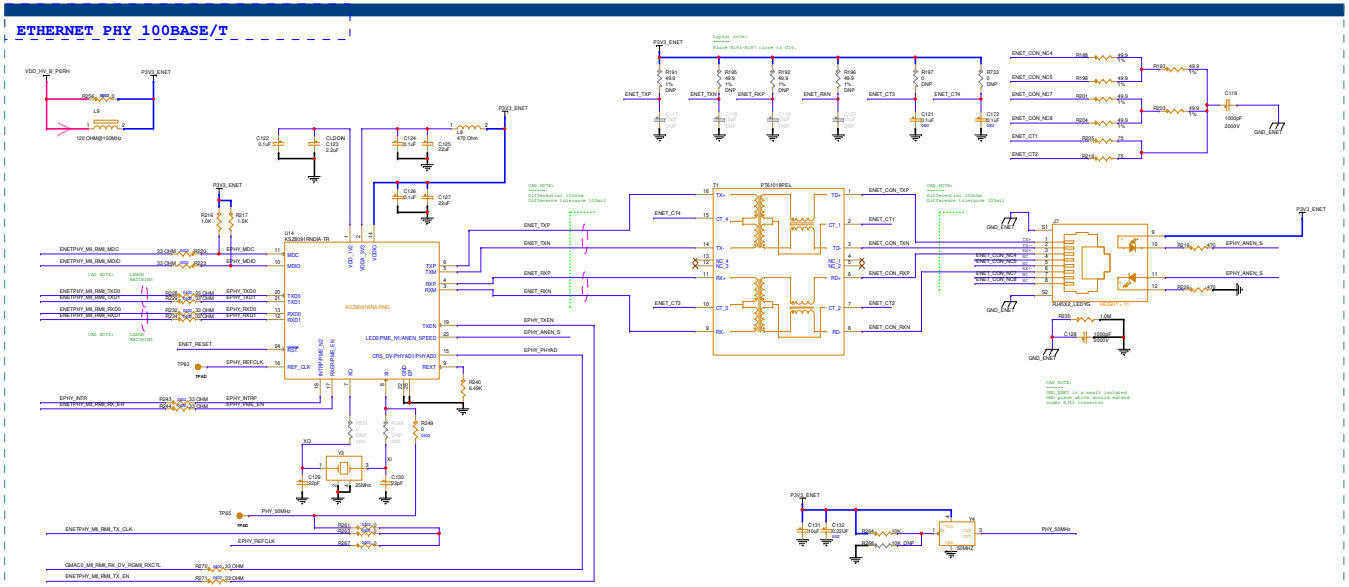
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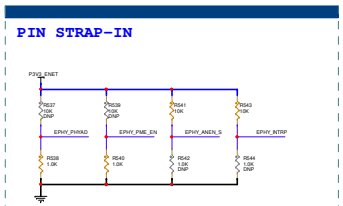
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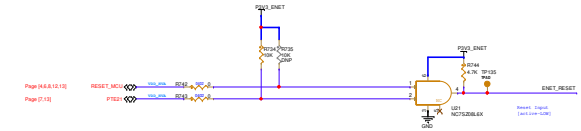
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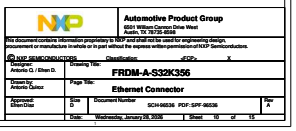
Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356	
Drawn by: Antonio Quiroz	Page Title: CAN/LIN PHY	
Approved: Efren Diaz	Size: C	Document Number: SCH-96536 PDF: SPF-96536 Rev: A
	Date: Wednesday, January 28, 2026	Sheet 9 of 15



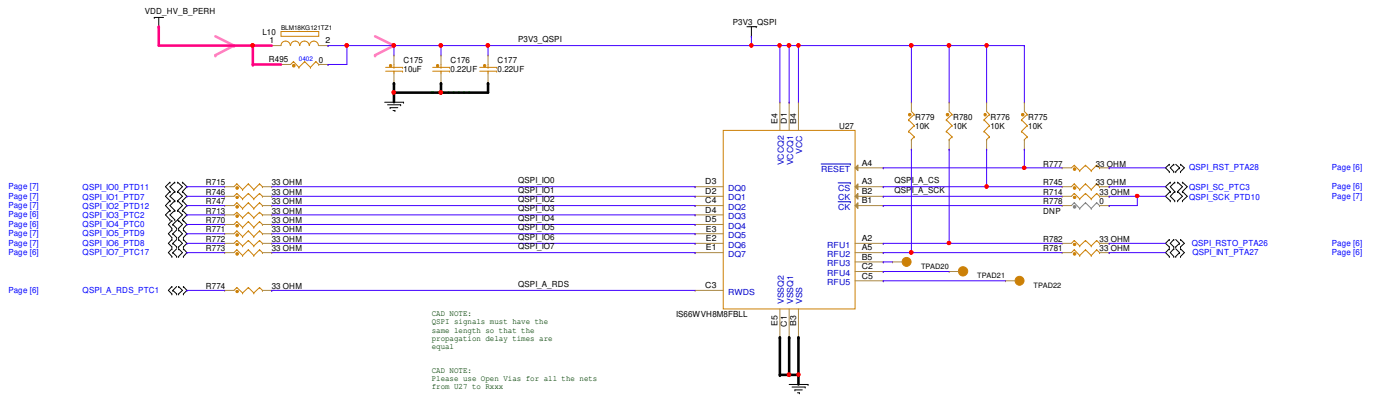

ETHERNET PHY 100BASE/T


PIN STRAP-IN

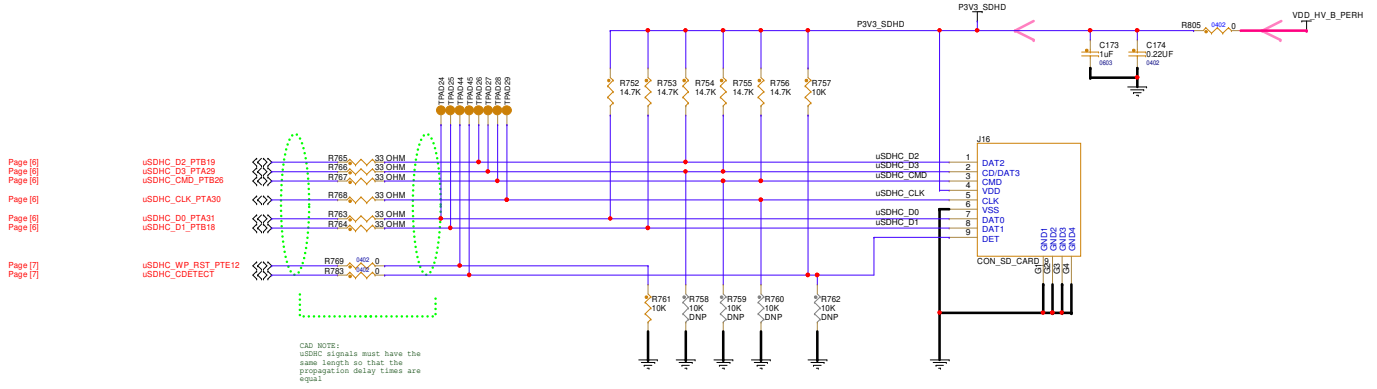



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FRDM-A-S32K356
Ethernet Connector
SCH-96536
PDF:SPF-96536
Wednesday, January 28, 2026

HyperRAM



SD Card



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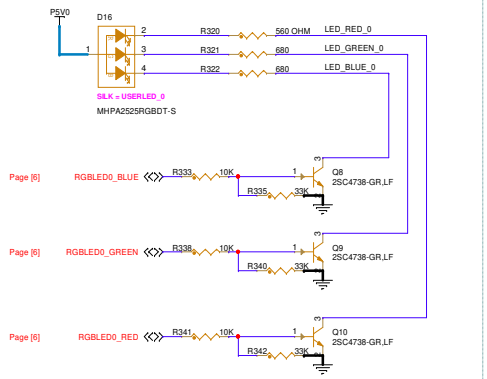
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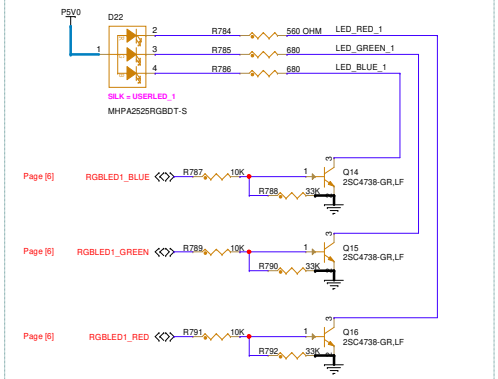
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Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356
Drawn by: Antonio Quiroz	Page Title: SD CARD / HyperRAM
Approved: Efren Diaz	Size C; Document Number SCH-96536 PDF: SPF-96536; Rev A
	Date: Wednesday, January 28, 2026; Sheet 11 of 15

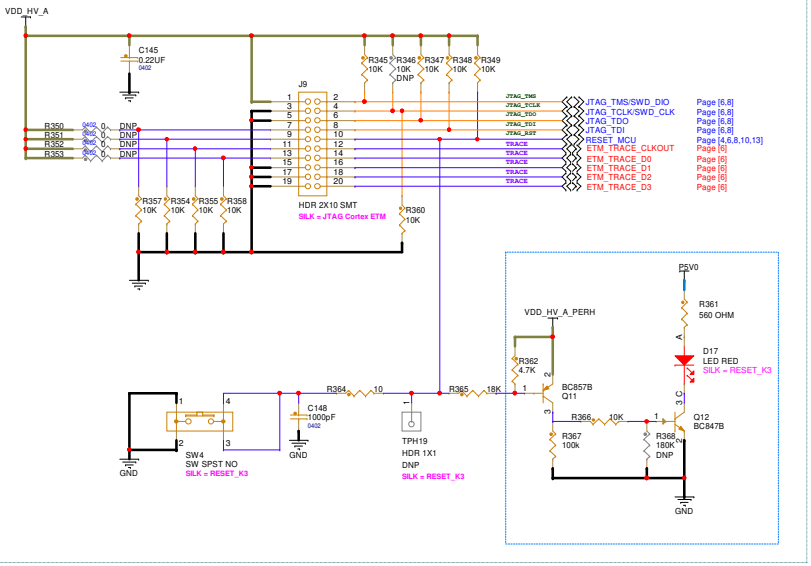
User RGB LED 0



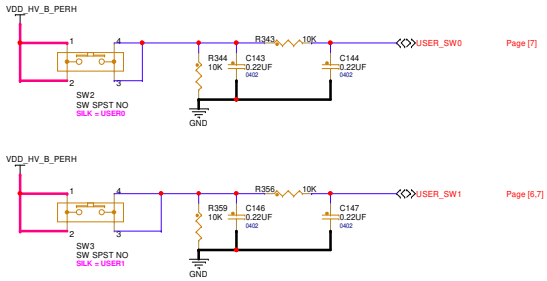
User RGB LED 1



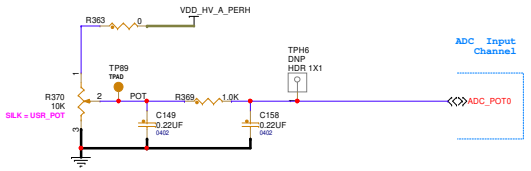
20-pin Cortex Debug D ETM connector



User Push Button - Active to HIGH/LOW



ADC Potentiometer



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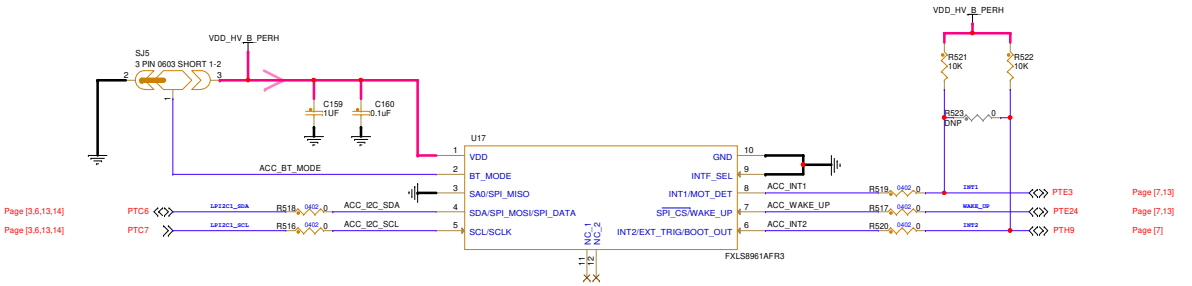
Automotive Product Group 6501 William Cannon Drive West Austin, TX 78735-8598		
Designer: Antonio Q. / Efren D.	Classification: <FCP> X	
Drawn by: Antonio Quiroz	Drawing Title: FRDM-A-S32K356	
Approved: Efren Diaz	Page Title: JTAG / PERIPHERALS	
	Size C / Document Number SCH-96536 PDF: SPF-96536	Rev A
	Date: Wednesday, January 28, 2026	Sheet 12 of 15



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Designer: Antonio G. / Efren D.	Drawing Title: FRDM-A-S32K356
Drawn by: Antonio Cuevas	Page Title: Arduino Shield
Approved: Efren Diaz	Size: D; Document Number: SCH-98536 PDF:SPF-98536; Rev: A
	Date: Wednesday, January 29, 2020; Sheet 13 of 15

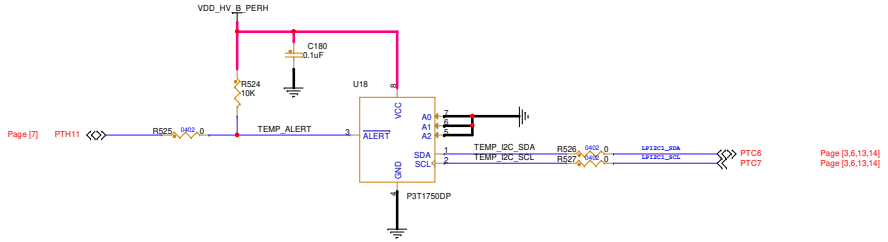
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ACCEL SENSOR



I2C ADDRESS: WR:30 RD:31

TEMP SENSOR



I2C ADDRESS: 0X90

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Designer: Antonio Q. / Efren D.	Drawing Title: FRDM-A-S32K356	
Drawn by: Antonio Quiroz	Page Title: Sensors	
Approved: Efren Diaz	Size: C	Document Number: SCH-96536 PDF: SPF-96536 Rev: A
	Date: Wednesday, January 28, 2026	Sheet 14 of 15

JUMPER Table

REF DES	JUMPER(DEFAULT)	PAGE NAME
JP11	1-2	04. FS26
JP4,JP5	1-2	05. MCU PWR
J12	OPEN	08. OPENSDA K26
J17	1-2	13. ARDUINO HDRS

DNP Table

REF DES	ASSY_OPT	PAGE NAME
R88,R123,R128,R448,R449,R455, R456,R457,R458,R502,R730, TPH23	DNP	03. USB TYPE C/PD
C1,C9,D1,J10,R4,R23,R42,R529, R732,R803,TPH2,TPH11	DNP	04. FS26
C62,C63,C64,Q5,R47,TPH3	DNP	05. MCU PWR
R50,R76	DNP	06. MCU PORTS1
R79	DNP	07. MCU PORTS2
C99,J12,R107,R115,R116,R117, R118,R138,R150,R153,TPH10	DNP	08. OPENSDA K26
C104,C105,C113,C114,C178, C179,R167,R168,R170,R180, R181,R182,TPH24	DNP	09. CAN/LIN PHY
C117,C118,C119,C120,R184, R185,R186,R191,R192,R195, R196,R197,R208,R209,R221, R222,R238,R239,R248,R251, R254,R255,R268,R537,R539, R542,R544,R733,R735	DNP	10. ETHERNET 10/100BASET
R758,R759,R760,R762,R778	DNP	11. SDCARD/HyperRAM
R346,R350,R351,R352,R353, R368,TPH6,TPH19	DNP	12. JTAG / PERIPHERALS
JB1,JB2,JB3,JB4,JB5,JB6,R398, R399,R400,R413,R414,R416, R418,R420,R421,R422,R430, R433,R435,R750,R751	DNP	13. ARDUINO HDRS
R523	DNP	14. SENSORS

SWITCH Table

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW5	1-2 / 4-5	03. USB TYPE C/PD

Shunt Table

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ4	1-2	05. MCU PWR
SJ1,SJ2	1-2	08. OPENSDA K26
SJ5	1-2	14. SENSORS

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